

MID EXAM MACHINE

Multiple choices

Choose the correct answer.

1- The open circuit test and short circuit test are performed on a transformer to determine the:

- a. Equivalent circuit of transformer
- b. Voltage regulation of transformer
- c. Efficiency of transformer
- d. All the above
- e. None of the above

2- Which of the following loss varies with the load:

- a. Copper loss
- b. Hysteresis loss
- c. Eddy current loss
- d. All losses in transformer vary with load

3- On ideal transformer draw current with no load connected to secondary winding:

a. False

b. True

4- On open circuit transformer test, voltage at rated frequency is applied to:

a. HV side

b. LV side

c. LV and HV side

d. LV or HV side

5- Transformer open circuited test, input power consists of core losses in transformer:

a. False

b. True

6- The primary and secondary winding of transformer have linked each other by:

a. Mutual conduction

b. Mutual induction

c. Not linked at all

d. Self-induction

7- Wings present in an autotransformer are:

a. 1

b. 2

c. 3

d. 4

8- Transformer transform:

a. Current

b. Voltage (Check for this)

c. Power

d. Frequency

9- The property of the material which oppose the creation of magnetic flux it's called:

a. Reluctivity

b. Permeability (Check for this)

c. Magnetomotive force

d. Reluctance

10- Below is the circuit diagram for delta-delta transformer have a transformation ratio 'k'.

The current flowing in the windings of phase r-y in the below diagrams is:

a. $k \cdot i / 1.73$

b. $i / k \cdot 1.73$

c. $1.37 \cdot k \cdot i$

d. $3 \cdot k \cdot i$

11- Secondary current of a step-down transformer is:

a. Lower than primary current

b. Higher than primary current

c. Equal to primary current

d. Double than primary current

12- The core of transformer must have:

- a. Low reluctance
- b. High reluctance
- c. Low resistance
- d. high resistance

13- An ideal transformer is one which has:

- a. No winding resistance
- b. No leakage reactance
- c. No losses
- d. All of above

14- Iron loss in transformer can be determined by:

- a. Short circuit test
- b. Open circuit test
- c. Both short circuit and open circuit tests
- d. None of the above

15- The law describing the electromotive force induced in a conductor placed in varying magnetic field is:

- a. Ohm's law**
- b. Gauss's law**
- c. Faraday's law**
- d. Ampere's law**

16- The unit of the magnetic flux is:

- a. Henry**
- b. Webber**
- c. Tesla**
- d. Ampere/meter**

17- A constant current flow in a horizontal wire in the plane of the paper from east to west as shown in figure.

The direction of magnetic field at a point will be north to south:

- a. Directly above the wire**
- b. Directly below the wire**

c. In the plane of the paper, In the north side of the wire.

d. In the plane of the paper, In the south side of the wire.

18- The electromotive force induced in a conductor of length l moving at velocity V in a magnetic field having a flux density B is:

a. $e_{ind} = (V \times B) \times l$

b. $e_{ind} = (V \cdot B) \cdot l$

c. $e_{ind} = (V \cdot B) \times l$

d. $e_{ind} = (V \times B) \cdot t$

19- The force F on a current carrying conductor of length l and current I , placed perpendicular in a uniform magnetic field of flux density B is:

a. $F = l/B I$

b. $F = B l I$

c. $F = B l I$

d. $F = l/B I$

20- In the power systems, the auto transformer is preferred over the ordinary two winding transformer due to:

- a. Safety of operation
- b. Availability of taps
- c. Material saving
- d. All the mentioned

21- The full-load copper loss of a transformer is 2000W, at half load, the copper loss will be:

- a. 400W
- b. 2000W
- c. 1000W
- d. 500W

22- A Transformer with primary winding having 100 turns, secondary winding having 500 turns, if 200V is applied to the primary winding, the voltage at secondary winding will be:

- a. 2000V
- b. 1000V
- c. 500V
- d. 100V

23- A Transformer with primary winding having 100 turns, secondary winding having 500 turns, if 200A is flowing into the primary winding, the current out of the secondary winding will be:

- a. 40A
- b. 100A
- c. 500A
- d. 1000A

24- The winding of the transformer that has less cross-sectional area of the conductor is:

- a. Primary winding
- b. Secondary winding
- c. High voltage winding
- d. Low voltage winding

25- If the rated speed of a DC motor is 1200RPM and the rated torque is 40N.m, the rated power of this motor is:

- a. 30KW
- b. 5KW

c. 3KW

d. 0.5KW

26- In the work done by a motor is 1000J, while its torque is 10N.m,
The number of complete revolutions done by shaft of this motor is:

a. 100π

b. 50π

c. $50/\pi$

d. $100/\pi$

27- In the field winding, of a lightly loaded DC shunt motor, gets
opened the:

a. Motor will stop

b. Motor speed will increase slightly

c. Motor speed will decrease slightly

d. Motor will pick up very high speed

28- If a DC motor have a full-load speed of 750RPM and
speed-regulation of 10%, the no-load speed of this motor:

a. 825RPM

b. 800RPM

c. 750RPM

d. 675RPM

29- In the separately excited DC motor, the relation between the motor speed n , and the field current I_f is:

a. n is directly proportional to I

b. n is directly proportional to the square of I_f

c. n is inversely proportional to I_f

d. n is inversely proportional to the square of I

30- The magnetic reluctance of a core of ferromagnetic material with relative of 4000, a mean path length of 100cm and the cross-sectional area of 150cm² is:

a. 0.01667 A.turn/Wb

b. 2500 A.turn/Wb

c. 8750 A.turn/Wb

d. 13263 A.turn/Wb

31- A core of ferromagnetic material has a cross-sectional area of 100cm^2 , if the magnetic flux through this core is 0.01Wb , the magnetic flux density within this core is:

- a. 0.1T
- b. 100T
- c. 10T
- d. 1.0T

32- A 3-phase transformer has its primary connected in delta and secondary in star, the secondary to primary turns ratio per phase is 5, What would be the secondary voltage if the primary voltage is 400V :

- a. 80V
- b. 3364V
- c. 2000V
- d. 138V
- e. None of the above

33- Open circuit test on transformers is conducted so as to get:

- a. Hysteresis losses
- b. Copper losses
- c. Core losses
- d. Eddy current losses

34- The property of material which opposes the creation of magnetic flux in it is known as

- a) Permeance b) Magnetomotive force c) Permittivity d) Reluctance

35- Point out the wrong statement. Magnetic leakage is undesirable in electrical machine because it

- a) Elevates their efficiency b) Lower their power efficiency c) Increase their cost of manufacture d) Lead to their increased weight e) Produce fringing

36- For time varying currents, the field or waves will be

- a) Electric b) Magneto static c) Electromagnetic d) Magnetical

The induced voltage will oppose the flux producing it

- a) True b) False

37- Calculate the emf when a coil of 100 turns is subjected to a flux rate of 0.2 tesla/sec

- a) 3 b) 20 c) -20 d) -200

38- Find the force due to a current element of length 2cm and flux density of 1.2 tesla. The current through the element will be 5A

- a) 0.12 N b) 1.2 N c) 14 N d) 1.4 N

39- Which of the following statements is true? a) $\mathbf{E}$ is the cross product of $\mathbf{v}$ and $\mathbf{B}$ b) $\mathbf{B}$ is the cross product of $\mathbf{v}$ and $\mathbf{E}$ c) $\mathbf{E}$ is the dot product of $\mathbf{v}$ and $\mathbf{B}$ d) $\mathbf{B}$ is the dot product of $\mathbf{v}$ and $\mathbf{I}$

Which parameter does not change during transformation action in transformer?

- a) Current b) Voltage c) Power d) Frequency e) Both A and B
f) None of the above

40- The primary and secondary winding of transformer are linked each other by a) conduction b) mutual induction c) not linked at all d) self induction

The power factor is the ratio of _____ power to the _____ power

- a) average, apparent b) apparent, reactive c) reactive, average d) apparent, true average

41- A transformer transforms

- a. Current b. Voltage c. Power d. Frequency

42- The primary and secondary winding of transformer are linked each other by

A conduction B mutual induction C not linked at all. D self induction

43- Which parameter does not change during transformation action in transformer?

A Current B Voltage C Frequency D both A and B E None of the above

44- The core of transformer must have

A low reluctance. B high resistance. C high reluctance D low resistance.

45- Which of the following losses varies with load in transformer?

A Copper loss. B Hysteresis loss. C Eddy current loss. D All losses in transformer vary with load.

46- A transformer transform

a. Current b. Voltage c. Power d. Frequency

47- The friction losses in Real Transformers are _____ ?

a) 5% b) 25% c) 50% d) Different from one transformer to other
e) None of the above

Problems

1

Q2) Solve and show your work (8 Marks)

Consider the linear DC machine as shown, with $R = 0.5 \text{ ohm}$, $B = 0.25 \text{ T}$, and $L = 2 \text{ m}$, with no load study state velocity of 300 m/s

$$V_{ss} = 300 \quad e_{\text{ind}} = VB \quad V_{ss} - e_{\text{ind}} = VB / BL$$

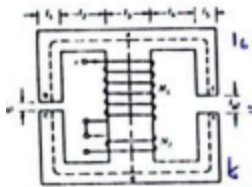
- Find e_{ind} and V_B
- Direction of the bar movement at no load, and e_{ind}
- I_{max}
- Magnitude of external force required to charge battery with 5 volt
- In (d), how much power the battery consuming or producing

2

Q3) Solve and show your work (10 Marks)

In the magnetic circuit shown below provides flux in the two air gaps. The coils ($N_1=700$, $N_2=200$) are connected in series and carry current of 0.5 amperes. Neglect leakage flux, reluctance of the iron (infinite permeability) and fringing at the air gaps.

- Draw the equivalent circuit.
- Determine the flux and flux density in the air gaps.



3

For autotransformer

- What are the uses of an autotransformer?
- What is the difference between a power transformer and an autotransformer?
- Draw the step-down autotransformer circuit

4

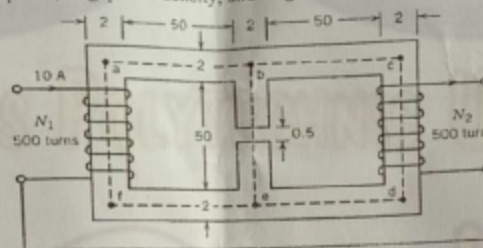
A single-phase transformer has 100 turns on the secondary and 1000 turns on primary. If transformer is connected two 110 volt supply while 3 kVA load connected to the secondary winding. Assuming ideal transformer, find,

- load impedance
- load voltage
- load impedance as referred to the primary

5

Solve and show your work

A magnetic circuit as shown below with relative permeability of the ferromagnetic material is 1200. Neglect magnetic leakage and fringing. All dimensions are symmetric in centimeters and magnetic material has square cross section area. If the length for b-e (only) is 51.5 cm, Find the airgap flux, airgap flux density, and magnetic field intensity in the air gap.



6

Q4) Solve and show your work

(7 Marks)

A core with three legs is shown. Its depth is 5 cm, and there are 100 turns on the leftmost leg. The relative permeability of the core can be assumed to be 2000 and constant. What flux exists in each of the three legs of the core? What is the flux density in each of the legs? Assume a percent increase in the effective area of the air gap due to hinging effects.

